Intermediate Report

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Repository: https://github.com/mrjones48/CVRP_NN_vs_ClarkeWright_vs_HarmonySearch

Coding

Mandy Jones implemented both the Nearest Neighbor Heuristic and the Clarke-Wright Savings Algorithm, including the shared utility functions for reading CVRPLIB data files and calculating Euclidean distances between customers. Both implementations were tested against all four benchmark instances (A-n32-k5, E-n51-k5, E-n101-k8, M-n200-k16) to confirm correct parsing, valid route construction, and distance calculations before being used in experimentation.

Tristen Martin implemented the Harmony Search algorithm, including the harmony memory structure, a deterministic memory-consideration and pitch-adjustment mechanism for generating new candidate solutions, and a local-search improvement step that refines each candidate through adjacent customer swaps.

Results Generation

A Python script, run_experiments.py, was developed to automate testing across all three algorithms and all four benchmark instances. The script runs each algorithm ten times per instance, records the resulting route distance and runtime for each trial, calculates the average runtime and the percentage gap from the best-known CVRPLIB solution, and outputs all results into a CSV file (located in results/all_results.csv).

Decisions Took Regarding the Experimentation

Regarding the research questions, I took out the ant colony question as it is no longer relevant. The requirements asked for "at least 3 research questions," so I did not replace the question. Our questions have a focus on resource trade-offs as asked; therefore, a replacement is not needed.

I rephrased "solution quality" to "percentage gap from the best-known CVRPLIB solution" as suggested. This led to some alterations to the wording in research questions.

Question 3 of the original research questions was switched out because we did not focus on varying vehicle capacity as an experimental variable. All instances were tested using their original, fixed capacity values rather than a range of capacities, so a question specifically about capacity's effect could not be meaningfully answered with our current experimental design.

The documentation for the first checkpoint mentioned that the input size ranged from 25 - 200 customers. Based on data resources used, the closest match is 31 - 199 customers, and that is what we used.

Experience With the Project

Communication among team members was limited for much of this checkpoint, with most coordination occurring close to the submission deadline despite earlier attempts to initiate discussion about task division. As a result, the team had difficulty reaching early agreement on how to divide the workload, which delayed the start of implementation for some components.

There was also some confusion about the scope of certain tasks. One team member expressed interest in handling data preparation but was unable to specify what that task would involve; ultimately, no separate data preparation was necessary, since the benchmark datasets used were already in a usable format. One team member did not respond to communication attempts or contribute to this checkpoint or the previous one.

Given these circumstances, the contribution percentages below reflect actual work completed on this checkpoint. The team plans to establish clearer task assignments and more consistent check-ins earlier in the next project phase to address these communication gaps.

Percentages of team contribution for this project checkpoint:

Mandy Jones: 80%

Tristen Martin: 20%

Tomu Yamashita: 0%

Charles Davis: 0%